```
harivasan@Harivasan:~/PES1UG24CS572-pes-vcs$ make all
gcc -Wall -Wextra -O2 -c object.c -o object.o
gcc -o pes object.o tree.o index.o commit.o pes.o -lcrypto
gcc -o test_objects test_objects.o object.o -lcrypto
gcc -o test_tree test_tree.o object.o tree.o -lcrypto
harivasan@Harivasan:~/PES1UG24CS572-pes-vcs$ ./test_objects
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e
5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed022
1bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
harivasan@Harivasan:~/PES1UG24CS572-pes-vcs$
```

Screenshot 1A: Output showing all tests passing

Screenshot 1B: showing the sharded directory structure.

```
harivasan@Harivasan:~/PES1UG24CS572-pes-vcs$ ./test_objects
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e
5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed022
1bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
harivasan@Harivasan:~/PES1UG24CS572-pes-vcs$ find .pes/objects -type f | sor
t
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f4
76
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566f
cf
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc
12
harivasan@Harivasan:~/PES1UG24CS572-pes-vcs$
```

creenshot 2A: Output showing all tests passing

```
harivasan@Harivasan:~/PES1UG24CS572-pes-vcs$ gcc -Wall -Wextra -O2 test_tree
.c object.o tree.o index.o -o test_tree -lcrypto && ./test_tree
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
harivasan@Harivasan:~/PES1UG24CS572-pes-vcs$
```

Screenshot 2B: Pick a tree object from to show the raw binary format.

```
harivasan@Harivasan:~/PES1UG24CS572-pes-vcs$ # If the tree hash in the commi
t starts with f4:
xxd .pes/objects/f4/e3b312fbd42e3d1a28a6193e4724ff95a376e21b7c845cd52e741362
c7cc78
00000000: 626c 6f62 2031 3000 4865 6c6c 6f20 5045  blob 10.Hello PE
00000010: 530a                                     S.
harivasan@Harivasan:~/PES1UG24CS572-pes-vcs$
```

Screenshot 3A:

```
harivasan@Harivasan:~/PES1UG24CS572-pes-vcs$ rm -rf .pes
harivasan@Harivasan:~/PES1UG24CS572-pes-vcs$ ./pes init
Initialized empty PES repository in .pes/
harivasan@Harivasan:~/PES1UG24CS572-pes-vcs$ echo "hello" > file1.txt
echo "world" > file2.tx
harivasan@Harivasan:~/PES1UG24CS572-pes-vcs$ echo "hello" > file1.txt
harivasan@Harivasan:~/PES1UG24CS572-pes-vcs$ echo "world" > file2.txt
harivasan@Harivasan:~/PES1UG24CS572-pes-vcs$ ./pes add file1.txt file2.txt
harivasan@Harivasan:~/PES1UG24CS572-pes-vcs$ ./pes status
Staged changes:
  staged:     file1.txt
  staged:     file2.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  untracked:  index.c
  untracked:  index.h
  untracked:  pes.h
  untracked:  commit.c
  untracked:  test.txt
  untracked:  test_sequence.sh
  untracked:  test_objects.c
  untracked:  Makefile
  untracked:  test_tree.c
  untracked:  object.c
  untracked:  test_tree
  untracked:  commit.h
  untracked:  tree.h
  untracked:  pes.c
  untracked:  file2.tx
  untracked:  .gitignore
  untracked:  README.md
  untracked:  test_objects
  untracked:  hello.txt
  untracked:  tree.c

harivasan@Harivasan:~/PES1UG24CS572-pes-vcs$
```

Screenshot 3B: `cat .pes/index` showing the text-format index with your entries.

```
harivasan@Harivasan:~/PES1UG24CS572-pes-vcs$ cat .pes/index
100644 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 1776
450166 6 file1.txt
100644 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 1776
450173 6 file2.txt
harivasan@Harivasan:~/PES1UG24CS572-pes-vcs$
```

Screenshot 4A: Output of `./pes/log` showing three commits with hashes, authors, timestamps, and messages.

```
harivasan@Harivasan:~/PES1UG24CS572-pes-vcs$ ./pes log
commit 99e953c72111370fd92051f05f84e51da7fe57310e04750476c7a73b7fcbbc07
Author: PES User <pes@localhost>
Date:   1776450465

    Add farewell

commit 63351acf3f8d237917323571e9bb70ca20c955c0b6e2b85b0e50cf885c230dba
Author: PES User <pes@localhost>
Date:   1776450311

    Add worldell

commit 5f154ed77592f6ca0672fb57a51426dd1aed37edcec2d15d2718e0962612f27b
Author: PES User <pes@localhost>
Date:   1776450310

    Initial committial commiQ

harivasan@Harivasan:~/PES1UG24CS572-pes-vcs$
```

Screenshot 4B: showing object store growth after three commits.

```
harivasan@Harivasan:~/PES1UG24CS572-pes-vcs$ find .pes/objects -type f | sor
t
.pes/objects/05/c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f
43
.pes/objects/2c/f8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9eb
b4
.pes/objects/46/013b9159237d7387548a83b99808954651307b22203bdea933db6c5c6f5e
6a
.pes/objects/5f/154ed77592f6ca0672fb57a51426dd1aed37edcec2d15d2718e0962612f2
7b
.pes/objects/63/351acf3f8d237917323571e9bb70ca20c955c0b6e2b85b0e50cf885c230d
ba
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c
39
.pes/objects/99/e953c72111370fd92051f05f84e51da7fe57310e04750476c7a73b7fcbbc
07
.pes/objects/d4/0446002cf982e16e57cc34ab58ffa357404fefefca660c1bc890a10e10a0
13
.pes/objects/e0/0c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f
7f
.pes/objects/e6/7ed66bcb4a708250a89f315d4d8bb92703343c84df834438880e13a68652
bc
.pes/objects/f6/c316350657dbe7f8e55bc230c7aa7942c1a96ca2fee0978db1eaf264c11d
14
harivasan@Harivasan:~/PES1UG24CS572-pes-vcs$
```

Screenshot 4C: showing reference chain

```
harivasan@Harivasan:~/PES1UG24CS572-pes-vcs$ cat .pes/refs/heads/main
cat .pes/HEAD
99e953c72111370fd92051f05f84e51da7fe57310e04750476c7a73b7fcbbc07
ref: refs/heads/main
harivasan@Harivasan:~/PES1UG24CS572-pes-vcs$
```

SCREENSHOT 10: showing final integration

```
harivasan@Harivasan:~/PES1UG24CS572-pes-vcs$ make test-integration
gcc -Wall -Wextra -O2 -c index.c -o index.o
index.c:156:17: warning: 'get_file_mode' defined but not used [-Wunused-function]
  156 | static uint32_t get_file_mode(const char *path) {
      |                 ^~~~~~~~~~~~~
gcc -Wall -Wextra -O2 -c commit.c -o commit.o
gcc -o pes object.o tree.o index.o commit.o pes.o -lcrypto
=== Running integration tests ===
bash test_sequence.sh
=== PES-VCS Integration Test ===

--- Repository Initialization ---
Initialized empty PES repository in .pes/
PASS: .pes/objects exists
PASS: .pes/refs/heads exists
PASS: .pes/HEAD exists

--- Staging Files ---
Status after add:
Staged changes:
  staged:     file.txt
  staged:     hello.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  (nothing to show)


--- First Commit ---
Committed: ff6be0246d5a... Initial commit

Log after first commit:
commit ff6be0246d5ad094456bbaf0fe3e7bff41e54811ef60fc66762a481772ba973e
Author: Harivasan <PES1UG24CS572>
Date:   1776451532

    Initial commit


--- Second Commit ---
Committed: 58de5357b34a... Update file.txt

--- Third Commit ---
Committed: d320297e3b35... Add farewell

--- Full History ---
commit d320297e3b35a61a08fcb899d685035156f69afcad81c9a8f9f57dd0c76f1f81
Author: Harivasan <PES1UG24CS572>
Date:   1776451532

    Add farewell

commit 58de5357b34a75c5f61d377ad7ad2cd845c1b10a18e75a2653c9ab97610550a5
Author: Harivasan <PES1UG24CS572>
Date:   1776451532

    Update file.txt

commit ff6be0246d5ad094456bbaf0fe3e7bff41e54811ef60fc66762a481772ba973e
Author: Harivasan <PES1UG24CS572>
Date:   1776451532

    Initial committial comQ


--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
d320297e3b35a61a08fcb899d685035156f69afcad81c9a8f9f57dd0c76f1f81

--- Object Store ---
Objects created:
10
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/49/0b8046918fb7be050b0af121cb6bab2541fe00d65de0173d62ddaa58ae7e0e
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e0d6695797533f50d
.pes/objects/58/de5357b34a75c5f61d377ad7ad2cd845c1b10a18e75a2653c9ab97610550a5
.pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa1974d901ce282f30e08de0ccd92
.pes/objects/bf/0fea3ff145b41b70b6eb091e9ab8ef2a2d60b26006dfc34b83421c68e6cd93
.pes/objects/cb/82a18c3817d6448b14d601807ffc642bd956469ec25672034cb90400d0ec2a
.pes/objects/d3/20297e3b35a61a08fcb899d685035156f69afcad81c9a8f9f57dd0c76f1f81
.pes/objects/db/07a1451ca9544dbf66d769b505377d765efae7adc6b97b75cc9d2b3b3da6ff
.pes/objects/ff/6be0246d5ad094456bbaf0fe3e7bff41e54811ef60fc66762a481772ba973e

=== All integration tests completed ===
harivasan@Harivasan:~/PES1UG24CS572-pes-vcs$
```